

RESEARCH ARTICLE

Intrinsic Defect-Induced Exotic Polarization Waves in Van der Waals Layered Topological Insulators

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Received: 11 December 2025 | **Revised:** 2 July 2026 | **Accepted:** 30 July 2026

Keywords: antisite defect | atomic-resolution electron microscopy | layered topological insulator | second harmonic generation | wavy polarization order

ABSTRACT

The engineering of inversion symmetry breaking in 2D materials, for example via interlayer twist-angle control, has initiated the ‘2D age’ due to its transformation on the landscape of fundamental research and technological advances. The typical paradigms include nonlinear optical response, topological polar vortices, superconductivity, etc. In parallel, intrinsic structural defects induced inversion symmetry breaking also serves as a lever to create emerging physical phenomena. Here, we report on discovery of antisite defects induced exotic polarization waves in bismuth telluride $(\text{Bi}_2)_m(\text{Bi}_2\text{Te}_3)_n$ topological insulator thin films by using atomic-resolution scanning transmission electron microscopy. Despite weak van der Waals (vdW) bonding, driven by stochastic antisite defects, a quasi-cycloidal polarization order is observed along normal direction of the vdW interface in Bi_2Te_3 phase. As metallic Bi-Bi bilayer is replaced by Bi-Te bilayer via regular Te_{Bi} antisite defect, hyperbolic polarization waves are observed at septuple layers (SLs) of the Bi_8Te_9 phase. Our first-principles calculations reveal that the face-shared unusual packing of acentric octahedral units and insufficient carrier screening are responsible for the polar metal states. Our findings provide a new degree of freedom to manipulate the physical properties of topological insulators (TIs) and their future device application.

1 | Introduction

Topological states of matter, such as topological insulators (TIs) [1, 2], magnetic skyrmions [3], hopfions [4] and topological ferroelectric metals [5], have attracted a great deal of attention for their revolutionary significance in unveiling the novel and emerging phenomena. Dictated by the nontrivial topological phases and encoded by Chern or winding numbers [6, 7], the topological materials usually exhibit exotic physical properties such as the quantum anomalous Hall effect [8, 9], topological magnetoelectric coupling [10], half-quantized surface

anomalous Hall conductivity [11], etc. When integrated into heterostructures, for example, via the interlayer twist angle, the newly formed elementary particles and functional entities offer a fertile ground for the design of next-generation electronic devices. This includes the Majorana zero modes stimulated by magnetic field for quantum computation [12], polarized light-induced transport of helical Dirac fermions for opto-spintronics [13], enhanced thermoelectric performance by reduced lattice thermal conductivity for energy conversion [14], and fatigue-resistant ferroelectrics for information storage [15].

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Protected by time-reversal symmetry to surface states, the TIs such as $(\text{Bi}_{1-x}\text{Sb}_x)_2\text{Te}_3$ are characterized by a nontrivial Z_2 topology for their bulk band structure, in which the strong spin-orbital coupling gives rise to massless Dirac-cone type dispersion and spin-momentum locking at the surface [16]. With regard to other members in the TI family, one may notice that almost all TIs, including topological crystalline insulators [17], magnetic TIs [18], and higher-order TIs [19], possess spatial inversion symmetry. Even the inversion symmetry is used as an indicator to search for topological superconductors [20] and Dirac semimetals [21]. Although the exotic topological surface states also exist on the surface of Weyl semimetals, they require breaking of either the time-reversal symmetry or the lattice inversion symmetry [22]. On this basis, studies have reported that external fields can drive a number of topological phase transitions, for example, metal-to-insulator transition in Ta_3S_2 Weyl semimetal [23], and magnetic quantum phase transition in Cr-doped $\text{Bi}_2(\text{Se}_x\text{Te}_{1-x})_3$ [24].

Lately, the engineering of inversion symmetry breaking [25] in 2D layered materials has spawned a plethora of intriguing polar phenomena, for example, ferroelectricity in AB-stacked h-BN bilayers [26] and in un-twisted MoS_2/WS_2 bilayers [27], topological polar vortices in twisted $\text{MoS}_2/\text{MoS}_2$ bilayer [28], etc. Alien from this, studies have reported that antisite-type point defects may break the inversion symmetry and trigger the nonpolar-polar phase transition [29–32], for example, in YFeO_3 , $(\text{Ca},\text{Sr})_3\text{Mn}_2\text{O}_7$, and SnTe . For the 2D Bi_2Se_3 and Bi_2Te_3 , the antisite defects have been found to be prevailing [33, 34] owing to their role in lowering the formation energy. However, the scenario of the potential polar order has never been explored, and its interaction with the spin-orbital coupling is pending as well, both of which play important roles in regulating the transport behavior of electrons and holes [35, 36]. Therefore, it becomes imperative to unravel the multi-order correlation so as to decipher the exotic phenomena such as the n-to-p type conduction transition mediated by temperature or sample thickness [29, 30], the impact of polarity on the topological surface states, etc.

In this work, the structural motifs of 2D layered Bi_2Te_3 and Bi_8Te_9 thin films, grown by molecular beam epitaxy on Si (111) substrates, are investigated by using atomic-resolution scanning transmission electron microscopy (STEM). The $(\text{Bi}_2)_m(\text{Bi}_2\text{Te}_3)_n$ is an infinitely adaptive series composed of Te-Bi-Te-Bi-Te quintuple-layer (QL) and Bi-Bi bilayer (BL) building blocks [37], where the interlayers are linked by van der Waals (vdW) force and the entire lattices are deemed to adopt a centrosymmetric $R\bar{3}m$ or $P\bar{3}m$ space group. Complying with the $\bar{3}m$ point group symmetry, each atomic layer follows an ABC stacking along the c axis, and each individual QL consisting of two face-sharing BiTe_6 octahedra is infinitely expanded in the ab plane (Figure 1a). Since both the QL and BL blocks exhibit a closed-shell feature [38], the ideal QLs and BLs can be arbitrarily stacked along c axis with the formation energy close to zero (Figure 1d).

2 | Symmetry and Symmetry Breaking

Despite having the inversion symmetry, the layer-dependent polarity, for example, in Bi_mTe_n ($m = 0, n = 1$), can still be analyzed through probing the geometric symmetry of the

tellurium octahedra that consist the QLs. Specifically, for an individual QL, each Bi atom in one octahedron coordinates with three Te1- and three Te2-site atoms, which are shared by three and six octahedra, respectively. According to the reported x-ray diffraction refinement data [39], one can readily see that the position of Bi atom deviates from the geometric center of the tellurium octahedral cage along z direction. This rationalizes consideration of octahedron-based “quantized” polarity of the QL, in which a pair of dipoles appears in an antiparallel manner with equal magnitude.

Depending on the amount of Bi-to-Te electron transfer, location of the negative charge centers (δ_i), as expressed by

$$\delta_i = \frac{\mu_1 z_{\text{Te1}} + 0.5\mu_2 z_{\text{Te2}}}{\mu_1 + 0.5\mu_2} \quad (1)$$

determines the dipole configuration in the QL. Here, the variables z_{Te1} and z_{Te2} are atomic sites of Te1 and Te2 atoms in one half of the QL, μ_1 and μ_2 are their valence states, respectively. With respect to the centrosymmetric position of Bi atom (δ_0), a $|\delta_i| > |\delta_0|$ relation (blue crosses) indicates that the octahedral pair exhibits a head-to-head dipole configuration (blue arrows). Otherwise, a $|\delta_i| < |\delta_0|$ relation (green crosses) yields a tail-to-tail configuration (green arrows) (Figure 1b and Experimental Section). A net polarity per QL unit is an average of that of the octahedral pair, that is, $P_{\text{QL}} = (P_{\text{Oct(i)}} + P_{\text{Oct(i+1)}})/2$. Here, the octahedral polarity $P_{\text{Oct(i)}}$ can be expressed as

$$P_{\text{Oct(i)}} = \frac{1}{V_i} \sum_j Z_{ij} R_{ij} \tau_{ij} \quad (2)$$

where Z_{ij} , R_{ij} , and τ_{ij} denote the effective charge, positions, and coordination of involved atoms in the i -th octahedron with a volume of V_i , respectively. For an ideal QL, the $P_{\text{Oct(i)}} = -P_{\text{Oct(i+1)}}$ relation holds true and thus the net polarity P_{QL} is null. Naturally, coincidence of the negative and positive charge centers at $|\delta_i| = |\delta_0|$ defines a critical transition point for the reversal of octahedral polarity.

With respect to the ideal QLs, the presence of Bi_{Te} and Te_{Bi} antisites and even atomic vacancies in either Bi-poor/Te-rich or Bi-rich/Te-poor conditions [33] can easily introduce local charge redistribution and distortion of the octahedra, inducing off-center shift of Bi atoms in the octahedra. As a result, the balanced polarity for the octahedral pair is broken, that is, $P_{\text{Oct(i)}} \neq -P_{\text{Oct(i+1)}}$, and a net polarity can be developed for a local QL unit. Since the octahedral distortion is impeded by the vdW gap, in principle, a long-range dipolar correlation is highly favored in the ab plane of the QLs, which could be analogous to ferromagnetic spin coupling between Mn^{2+} ions within each individual septuple layers (SLs) in MnBi_2Te_4 [40]. It should be noted that for the Bi_2Te_3 system without an additional source for charge screening [41], the net polarization tends to orient flexibly in the 3D space (Figure 1c). In contrast, although with weak covalent bonding to the QLs [38], only z -direction polarity is energetically preferred in compounds like Bi_8Te_9 owing to the presence of Bi-Bi BLs for charge screening (Figure 1e,f).

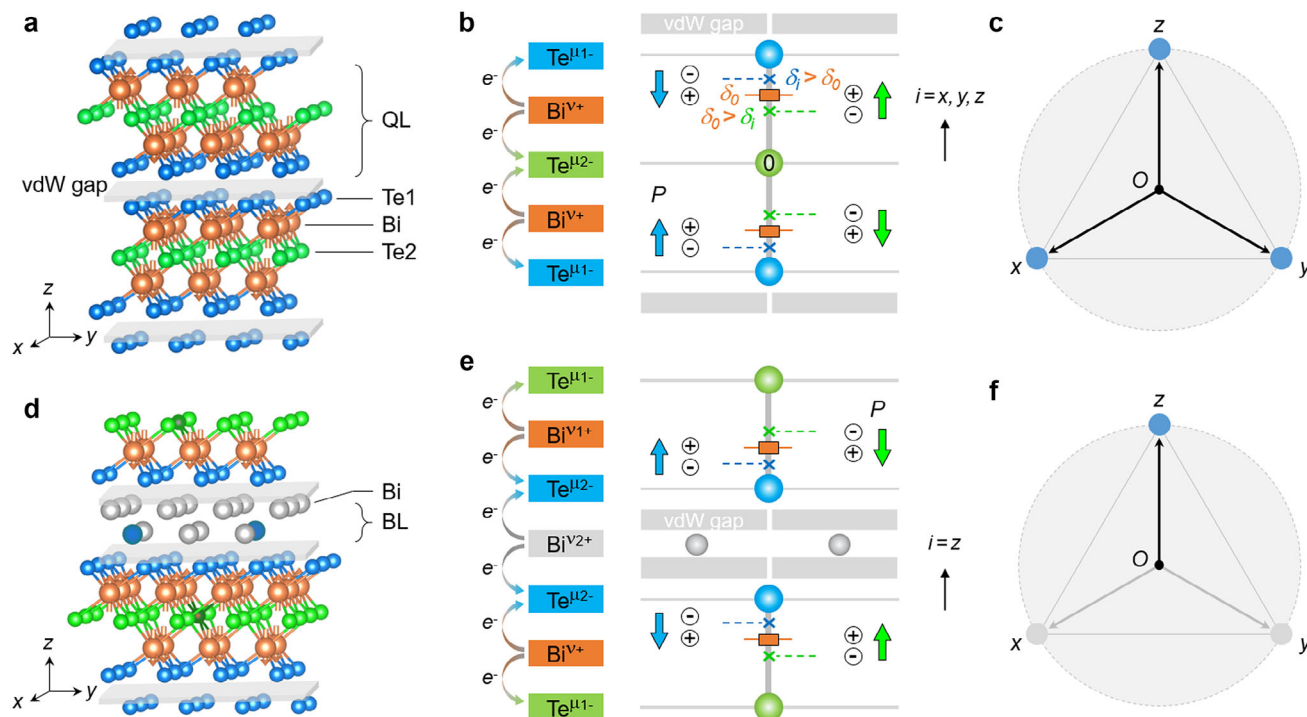


FIGURE 1 | Pair-based octahedral polarity and preferential orientation in $(\text{Bi}_2)_m(\text{Bi}_2\text{Te}_3)_n$. (a,d) Crystal structures of ideal QL stacking and QL-BL-QL stacking. The dark atoms in (d) denote either antisite defects such as Bi_{Te} , Te_{Bi} , or atomic vacancies. (b,e) Layer-wise charge transfer between Bi and Te layers along z axis and octahedral polarity (blue and green arrows) arising from separation of Bi-atom positive charge centre (orange rectangles at position of δ_0) from Te-atom negative charge centre (green and blue spheres at position of δ_i) within a QL and near a Bi-Bi BL, respectively. (c,f) Illustration of possible polarity orientation (black thick arrows) in QL-stacked Bi_2Te_3 along x, y, and z directions and preferential polarity orientation (black thick arrow) along z direction in -QL-BL-QL-stacked systems, respectively.

3 | Thin-Film Growth and Antisite Defects

To explore the polarity-related structural features, we grew high-quality Bi_2Te_3 and Bi_8Te_9 thin films on Te-terminated Si (111) substrates by using molecular beam epitaxy and cross-sectional lamellae specimens were prepared along [100] direction for electron microscopy investigation (see Experimental Section). At the microscopic scale, high-angle annular-dark-field (HAADF) STEM, with the image contrast proportional to the $Z^{1.6-2.0}$ (Z : atomic number) [42], was used to image and analyze their structural motifs. At medium resolution, our HAADF-STEM image reveals that instead of an ideal QL stacking along z direction, twin-like domains with a displacement vector of $\mathbf{R} = (\mathbf{a} + 2\mathbf{b})/3$ can be frequently seen in the Bi_2Te_3 films with a thickness around 15 nm (Figure 2a,b). In the Bi_8Te_9 sample, we clearly observe a twin-like boundary as well. In between the boundary and the Bi-Te/Si interface, the SLs arising from adjacency of neighboring BL-QL pairs can be identified from the modulation contrast and spacing in the inverse fast Fourier transform (IFFT) image, which forms the Bi_8Te_9 phase with $m = 1$ and $n = 3$ (Figures 2g,h and S1). Above this boundary, the ordered QL and SL stacking yields a Bi_5Te_6 phase with $m = 1$ and $n = 4$. Given the flexible QL-BL combination, this implies that the Bi-Te systems can be understood from the viewpoint of structural domains, analogous to ferroelastic domains in antiferroelectric materials [43].

To examine the chemical distribution information, energy-dispersive x-ray spectroscopy (EDS) was performed on the cross-sectional samples. From the atomically resolved Bi and Te maps, we see the presence of more Te_{Bi} -type antisite defects than the Bi_{Te} -type ones in the Bi_2Te_3 sample, which are marked by white and blue dashed circles on the EDS maps, respectively (Figure 2c-e). Specifically, the major Te_{Bi} -type defects are distributed in a random manner, and their mutual spacing is around 1–6 octahedral units among the QLs. In statistics, the atomic ratio is evaluated as $\text{Te}/\text{Bi} \approx 1.56 \pm 0.06$, slightly higher than the nominal ratio of 1.50, and the atomic percentage profile suggests that fractional Bi tends to appear at the vdW gap (Figure 2f).

As for the Bi_8Te_9 sample, although the BLs sandwiched by the QLs can be identified easily, the atom contrast in the HAADF image indicates that one layer is dominated by Bi, but the other layer tends to be comprised of mixed Bi and Te or even dominated by Te (Figure 2g-k). On the one hand, this is substantiated by observation of more regular occupation of Te at the Bi site in the BLs from the EDS map, where a few Bi_{Te} antisite defects can also be seen. On the other hand, this is supported by our simulated HAADF images based on models with varied Bi/Te ratio in the BLs (Figure S2). As a result, the SL can be regarded as a combination of one triple layer (TL) sandwiched by two Bi-Te-like BLs, whose layered structural feature can be discerned from the atomic arrangement, compositional profile and the Te/Bi ratio measured as 1.26 ± 0.06 (Figure 2l).

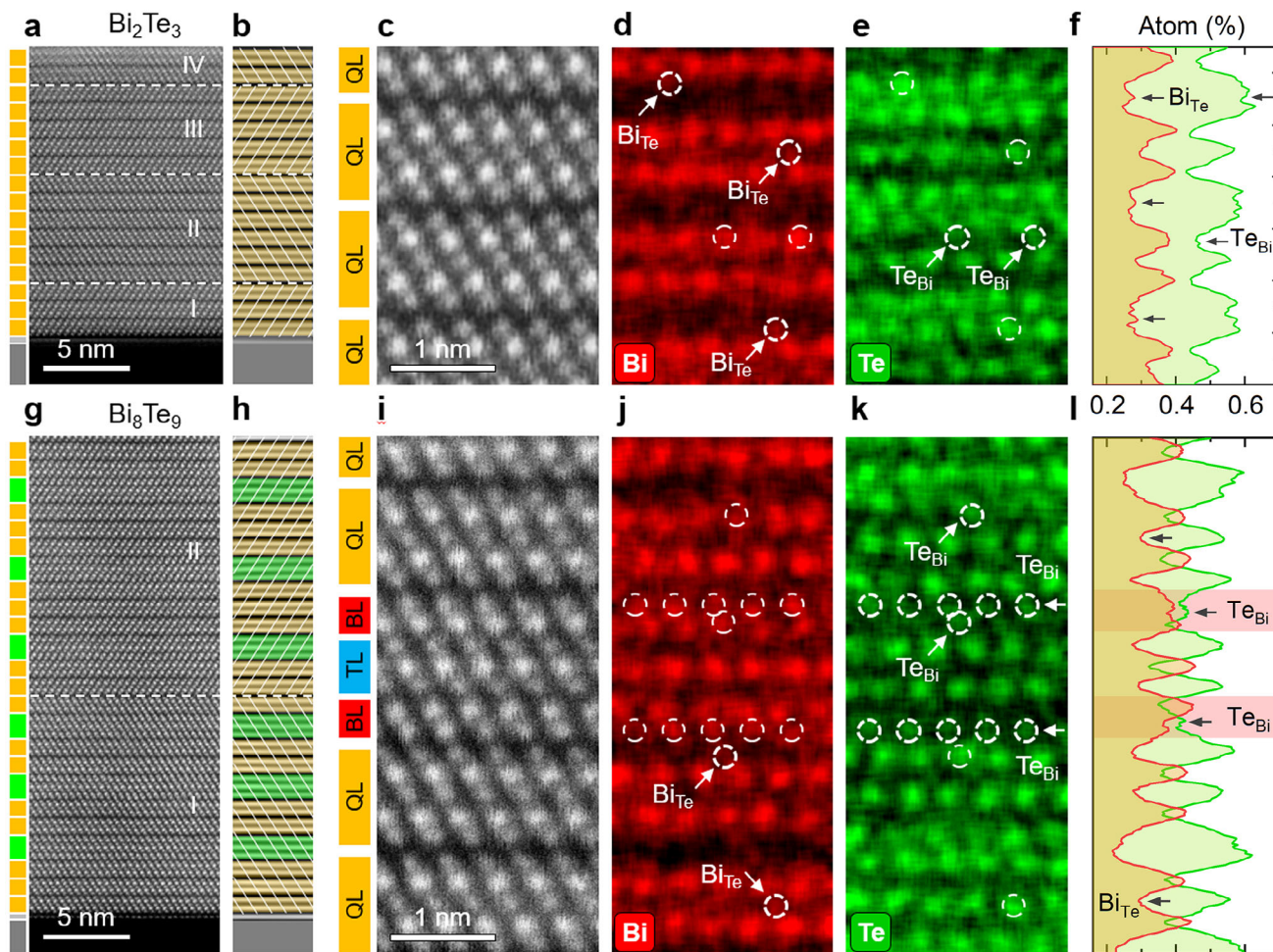


FIGURE 2 | Structural domains and atomic antisite defects. (a, g) HAADF-STEM images of Bi_2Te_3 and Bi_8Te_9 films on Si (111) substrates collected along [100] direction, respectively. (b, h) Corresponding inverse fast Fourier transform (IFFT) images used to highlight the stacking QLs (orange blocks), SLs (green blocks) and twin-like boundaries (white dashed lines) lying at (001) planes, respectively. (c–f) HAADF image of Bi_2Te_3 and corresponding elemental maps of Bi and Te and line profiles, respectively. (i–l) HAADF image of Bi_8Te_9 , corresponding elemental maps of Bi and Te and line profiles, respectively. The thick dashed circles denote the atomic antisite defects and the thin dashed circles denote the corresponding atomic positions, respectively.

4 | Exotic Wavy Polarization Order

Through measuring the positions of Bi and Te atoms via fitting the column peak intensities using 2D Gaussian functions, we quantified the structural parameters of the Bi_2Te_3 sample. Our measurement reveals that the vdW gap value, that is, z -direction inter-planar spacing (d_{IPS}), agrees well with the reported experimental values, with $d_{\text{IPS}} = 0.264 \pm 0.005$ nm [44]. With observation of additional interstitial atoms, we find that this value is increased to $d_{\text{IPS}} \approx 0.294$ nm at the twin-like boundary (Figure 3a–c). Referring to the Bi–Bi positions in the QLs, which are linearly fitted to evaluate the relative shifts of Te atoms, we find that the termination Te1 atoms in the QLs swing to the right or left along y direction near the vdW gaps by ~ 20 pm, while the central Te2 atoms are almost immobile.

When quantifying the displacements of Bi atoms (δ_{Biz} , δ_{Biy}) against centers of the Te octahedra (Figure 3d), one finds that they preserve an antiparallel shift along z direction. However,

the displacement magnitudes become unequal in the QLs and an overall net displacement, with an antiparallel configuration across the twin-like boundary, is observed along the normal direction of the vdW interface (Figures 3d and S3). Along y direction, a polar shift of Bi atom is also observed, which stays around $\delta_{\text{Biy}} \approx -5.0$ pm over a long distance. Based on the Bader charges, that is, $Z_{\text{Bi}} = 0.685$, $Z_{\text{Te1}} = -0.38$ and $Z_{\text{Te2}} = -0.61$ [45], we calculate the octahedral dipole moments and average the values of neighbouring ones. With observation of layer-dependent P_{Sz} and P_{Sy} net polarization, with $\langle |P_{\text{Sz}}| \rangle = 0.54$ and $\langle |P_{\text{Sy}}| \rangle = 0.78 \mu\text{C cm}^{-2}$, we find that the overall polarization vectors in 2D space exhibit a pronounced wavy feature, which can be nicely approximated by a quasi-cycloidal polar order (Figure 3e). The mathematical justification is presented in Figure S4. Interestingly, the polarization reversal is also observed in regions with ideal QL stacking, which therefore can be attributed to the random distribution of antisite defects (Figure 2). By overlapping the discrete polarization vectors per QL on the HAADF image (Figures 3a and S5a), one can see that the hidden polarization

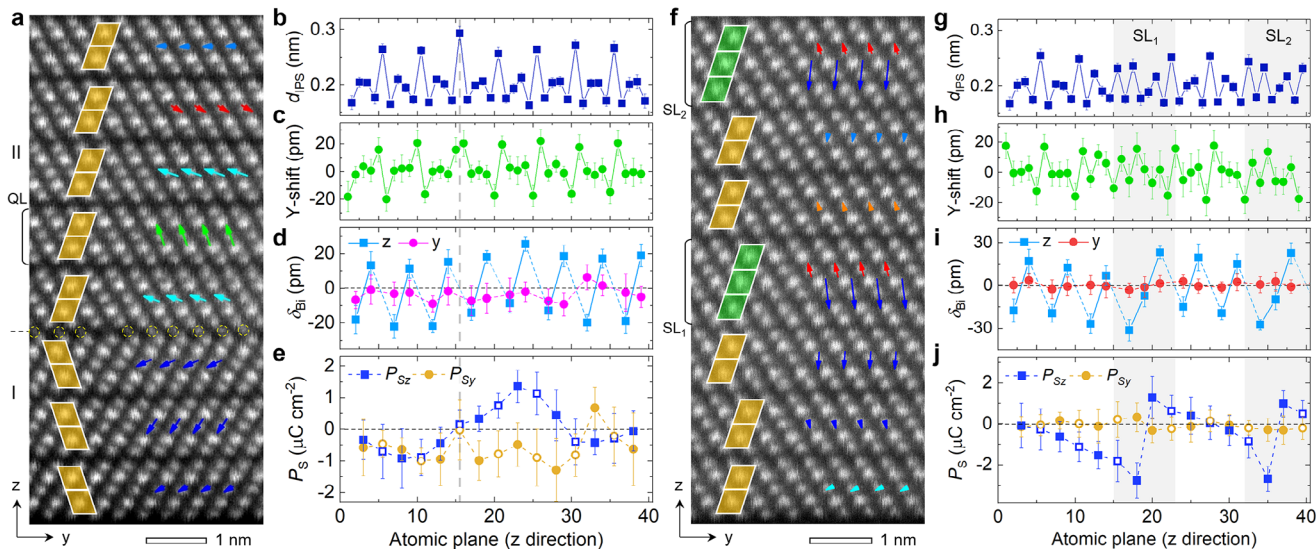


FIGURE 3 | Hidden wavy polarization orders. (a, f) HAADF-STEM image of Bi_2Te_3 and Bi_8Te_9 phase collected along [100] direction. The BiTe_6 octahedral units (orange and green shadows), QL-based 2D polarization vectors are overlaid on the images. The dashed line and circles mark the twin-like boundary the interstitial atoms. (b–e) Inter-planar spacing (d_{IPS}) (b), atomic y-direction shifts against the reference Bi columns in Te-Bi-Te-Bi-Te chains (c), polar shifts of Bi in the octahedra (δ_{ZBi} , δ_{YBi}) (d) and 2D layer-dependent discrete polarization (P_{Sz} , P_{Sy}) (e) for the Bi_2Te_3 phase, respectively. The dashed line denotes the twin-like domain boundary. (g–j) The d_{IPS} (g), atomic y-direction shifts (h), polar shifts of Bi (δ_{ZBi} , δ_{YBi}) (i) and 2D layer-dependent discrete polarization (P_{Sz} , P_{Sy}) (j) for the Bi_8Te_9 phase, respectively. The grey shadows mark the SL regions.

order forms a tail-to-tail geometry at the twin-like boundary, which indicates that the interstitial sites are occupied by Bi atoms due to preferential Te-Bi bonding and the requirement of hole screening.

In contrast, dramatic structural differences are observed in the Bi_8Te_9 sample. On the one hand, the vdW gap between adjacent QLs decreases to $d_{\text{IPS}} \approx 0.251 \pm 0.004$ nm, suggesting enhancement of the interlayer bonding between the building blocks. On the other hand, the SLs exhibit unique structural motifs, including the reduced vdW gap ($d_{\text{IPS}} \approx 0.233$ nm), changes in the -Bi-Te-Bi- interatomic spacing and shifts of Te1 and Te2 atoms along y direction inside the SLs (Figure 3f–h). Given the established Bi-Te-like BL feature, we designate the atom column types and calculate the octahedral and QL polarity according to the coordination marked by orange and green shadows (Figures 3f and S5b). Along z direction, one observes a gradual increase of imbalanced Bi shifts in the octahedral units (δ_{Biz}), which reaches the maximum asymmetry as approaching the first SL, with $P_{\text{Sz-min}} \approx -2.76 \mu\text{C cm}^{-2}$. Following this, the polarity suddenly reverses to the positive direction with $P_{\text{Sz-max}} \approx 1.28 \mu\text{C cm}^{-2}$. Such a hyperbolic polarization wave is repetitively observed in the subsequent SLs (Figure 3i,j), and the polarity asymmetry ($\Delta P_{\text{Sz}} = P_{\text{Sz-min}} + P_{\text{Sz-max}}$) gradually disappears as the SLs are farther away from the $\text{Bi}_8\text{Te}_9/\text{Si}$ interface (Figure S6). Distinct from the Bi_2Te_3 , the shift of Bi atom (δ_{Biy}) is nearly zero along y direction except at the polarity reversal point, whose net value is $P_{\text{Sy}} \approx 1.44 \mu\text{C cm}^{-2}$.

5 | Stochastic Antisite Induced Polarity and Persistent Topological States

To establish a clearer connection between theory and experiment, we constructed a series of Bi_2Te_3 supercells with sizes at $1 \times n$

$\times 12$ octahedral units. For a relatively simple case, we consider the $n = 2$ supercells that contain a pair of Te_{Bi} antisite defects with a spacing of $\Delta y = 2$ and $\Delta z = 1 \sim 6$, respectively. As can be seen, the polar octahedral distortion ($P_{\langle \text{Oct} \rangle}$) takes place in a near-spherical region with a radius of $2 \sim 3$ octahedral units (Figure S7). Associated with the presence and reversal of QL polarity ($P_{\langle \text{QL} \rangle}$) along z axis, net QL polarity along y direction can also be observed, which well agrees with the scenario presented in Figure 1c. On this basis, we constructed larger supercells ($n = 4, 6, 8$) that contain multiple pairs of $\text{Te}_{\text{Bi}}\text{-Te}_{\text{Bi}}$ antisite defects with an average spacing at ($\langle \Delta y \rangle = 2$, $\langle \Delta z \rangle = 1$), ($\langle \Delta y \rangle = 3$, $\langle \Delta z \rangle = 2$), and ($\langle \Delta y \rangle = 4$, $\langle \Delta z \rangle = 3$), respectively. By mapping the polar shifts of Bi in the octahedra, we find that the high-density situation (Figures 4a,b and S8), that is, $\langle \Delta y \rangle = 2$ and $\langle \Delta z \rangle = 1$, nicely reproduces our experimental data (Figures 2c–e and S5a). With increase of the defect-pair spacing, that is, from ($\langle \Delta y \rangle = 3$, $\langle \Delta z \rangle = 2$) to ($\langle \Delta y \rangle = 4$, $\langle \Delta z \rangle = 3$), one observes nearly linear reduction of the QL polarity along y and z direction due to the impaired dipolar correlation in 3D space (Figures 4d–f and S9, S10).

Compared with the ideally nonpolar Bi_2Te_3 phase, our first-principles calculations on $n = 2$ supercells reveal that consideration of one pair of antisite defects transforms the insulating bulk phase into semimetallic phases, for example, the $\text{Te}_{\text{Bi}}\text{-Te}_{\text{Bi}}$ spacing is $\Delta y = 2$, $\Delta z = 2$ and $\Delta y = 2$, $\Delta z = 6$ (Figure S11). Referring to this, we further calculated the bulk and surface-state band structures of larger supercells ($n = 4$), that is, $\langle \Delta y \rangle = 2$ and $\langle \Delta z \rangle = 1$, that contain the high-density antisite defects (Figure 4c). Associated with certain-degree compression of electronic bands toward flat ones, the Dirac cones manifested by crossing of two bands near the fermi level (E_{F}) demonstrate robustness of the topological surface states against the stochastic antisite defects. This scenario is consistent with the experimental findings

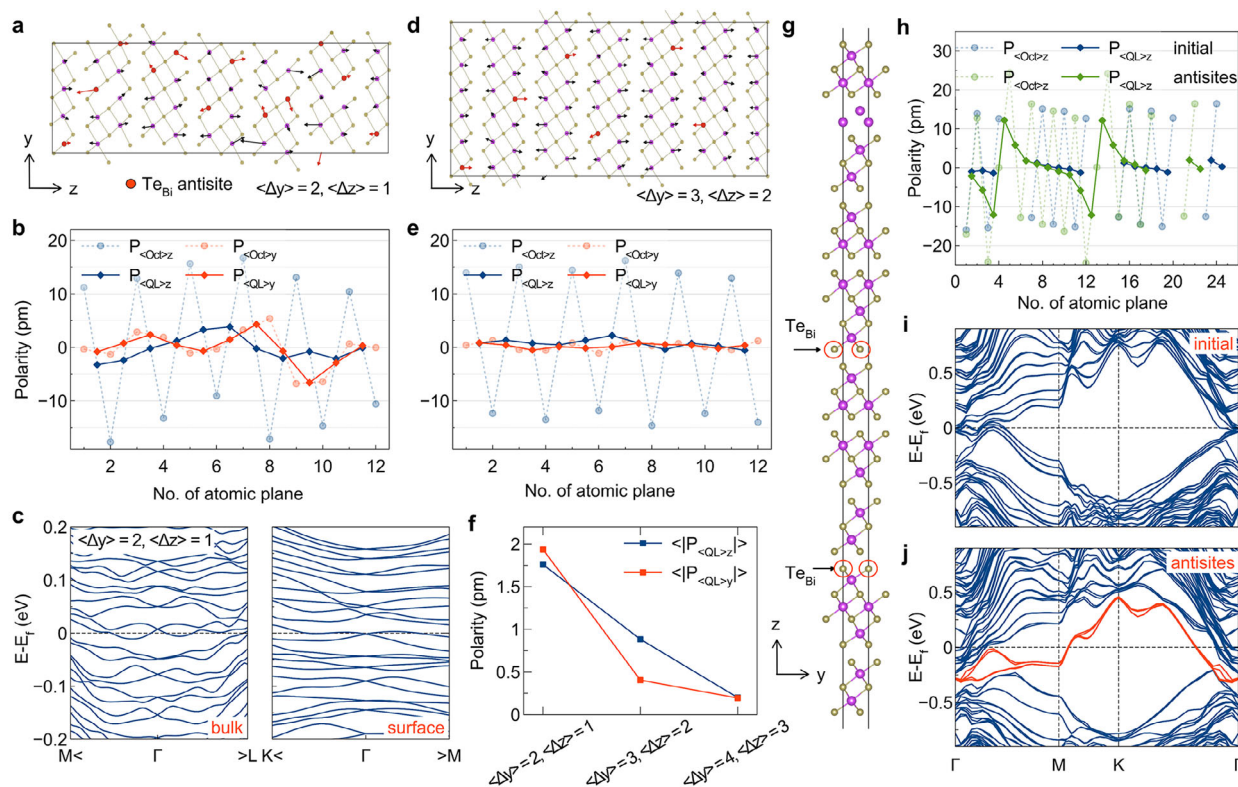


FIGURE 4 | Stochastic antisites induced dipolar correlation and persistent topological surface states. (a, d) Bi₂Te₃ supercells containing multiple pairs of Te_{Bi} antisite defects with an average spacing of (<Δy> = 2, <Δz> = 1) and (<Δy> = 3, <Δz> = 2), respectively. (b, e) Octahedral and QL polarity corresponding to (a) and (d), respectively. (c) Calculated bulk and surface-state band structures for (a) with high-density antisite defects. (f) Dependence of QL polarity on the average Te_{Bi}-Te_{Bi} spacing in 1 × n × 12 (n = 4, 6, 8) supercells. (g) A Bi₈Te₉ supercell containing two arrays of Te_{Bi} antisite defects with a spacing of Δz = 7 for surface state calculations, respectively. (h) Comparison of two-array antisites induced $P_{\langle \text{Oct} \rangle z}$ and $P_{\langle \text{QL} \rangle z}$ with that in the initial Bi₈Te₉ phase. (i, j) First-principles calculated band structures of the initial Bi₈Te₉ supercell (i) and the one containing two-array Te_{Bi} antisite defects, respectively. The antisite induced bands near E_F are highlighted by red lines. The SOC strength used in the calculations is $f_{\text{SOC}} = 1.0$. Compared with their topological surface states, their bulk-phase band structures are presented in Figures S11–S14.

reported in Bi(111)/Bi₂Te₃ system [46]. As anticipated, such a conclusion can be extended to supercells (n = 6 and 8) containing relatively low-density Te_{Bi} antisite defects.

For the Bi₈Te₉ phase, we started from an idealized periodic supercell and introduced one or two arrays of Te_{Bi}-type antisite defect in the supercells, where their positions and concentrations are varied along with converting the Bi-Bi to Bi-Te BL (Figures 4g and S12). Being consistent with the experimental result, the ordering of Te_{Bi} antisite defects leads to formation of quasi-1D polarity along z direction, that is, $P_{\langle \text{QL} \rangle z}$, where the dipoles form tail-to-tail configuration at the defect-locating (001) plane. The research outcome applies to supercells containing either the single- or two-array of Te_{Bi} antisite defects, where their bulk phases adopt a semimetallic state (Figures 4h and S13, S14). Their corresponding band structures show that the key topological feature—namely, band crossings at the Fermi level that generate metallic surface states with electron and hole pockets—remains preserved in all considered configurations (Figure 4i,j). These results demonstrate not only the robustness of topological attribute against the stochastic antisite defects but also their intimate correlation with the exotic wavy polar orders in the Bi₂Te₃ and Bi₈Te₉ phases.

6 | Confirmation of Dipolar Correlation by SHG

To further verify the antisite defect-induced inversion symmetry breaking and its correlation with dipolar correlation in 3D space, we performed optical second harmonic generation (SHG) experiments on a series of Bi₂Te₃ films with thickness $t = 5$ nm, 8 nm and 12 nm. Herein, the wavelength of the incident beam is $\lambda = 1064$ nm and the signal is detected along normal direction of the films (Figure 5a). Referring to the Si substrate, one finds that the SHG intensity at $\lambda = 532$ nm keeps increasing with increment of the film thickness from 5 nm to 12 nm (Figure 5b). This unambiguously proves that (1) the amount of random antisite defect increases with film thickness; (2) the antisite defects break the inversion symmetry, and the dipolar correlation is substantially enhanced with increase of the film thickness.

For the Bi₂Te₃ film with $t = 12$ nm, our measurement further reveals a six-lobed SHG pattern, which verifies a threefold rotational symmetry that complies with the C_{3v} point group (Figure 5c). Although the out-of-plane polarization component cannot be detected, the data confirm the macroscopic in-plane inversion symmetry breaking, which is consistent with the quasi-cycloidal wave observed in the Bi₂Te₃ phase (Figure 3a,e). By

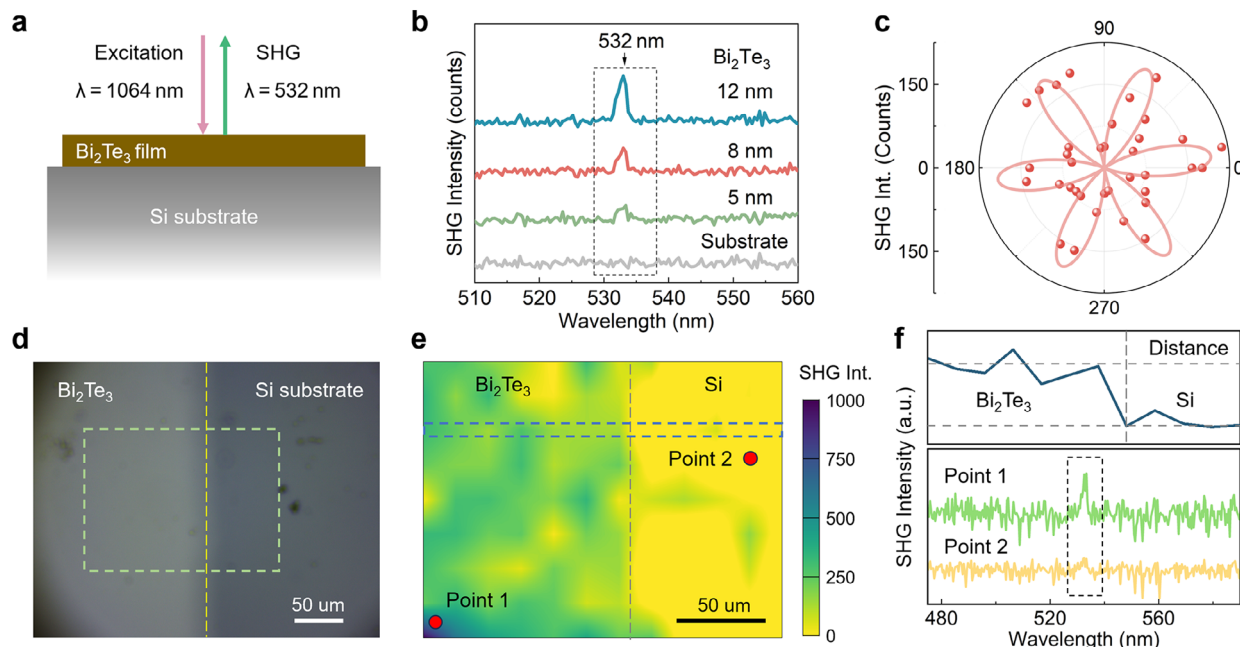


FIGURE 5 | SHG measurement on global polar responses in Bi_2Te_3 films. (a) Schematic diagram of the measurement setup. (b) SHG intensity as a function of different Bi_2Te_3 film thickness. The Si substrate is taken as a reference. (c, d) Angle-dependent polarization response along with its fit (solid line) and optical image of the Bi_2Te_3 film (thickness = 12 nm) on Si substrate, respectively. (e) SHG mapping of a local region marked by a green dashed box in (d). (f) SHG intensity profile and spectra corresponding to the blue dashed rectangle (up) and two red points 1 and 2 (bottom) in (e), respectively.

mapping the SHG signals of the Bi_2Te_3 film ($t = 12$ nm), one finds that the stochastic antisite defects lead to a thermodynamically stable phase (Figure 5d,e). This is substantiated by a SHG intensity profile extracted from a line region (marked by a blue dashed rectangle) and two SHG spectra (red points) taken from the Bi_2Te_3 sample and the Si substrate, respectively (Figure 5e,f).

7 | Outlook and Conclusion

Since the Bi-Te systems are surface metallic or semimetallic, implying a high density of free carriers, it becomes important to consider the screening of electrostatic fields by free carriers. Distinct from the ordinary ferroelectrics such as BaTiO_3 and PbTiO_3 , where the polarization excludes with the metallicity, it has been reported that packing of acentric polyhedral units, as evidenced in $\text{SrCaRu}_2\text{O}_6$ and Ni_2P [5, 47], can give rise to coexistence of polar order with metallic state. By breaking the inversion symmetry via stochastic antisite defects, the unusual face-shared BiTe_6 octahedra provide a favorable condition for the polar metal state. One is that the off-center shifts of Bi atoms inside the BiTe_6 cages are responsible for generating the local dipoles, which link with each other and establish the dipolar correlation in 3D space. Meanwhile, the Bi-Bi bonding of neighboring BiTe_6 cage leads to the metallicity.

Furthermore, we estimate the screening length (λ_{FT}) using the Thomas-Fermi screening theory [48], that is, $\lambda_{TF} = \frac{1}{k_{TF}} = \frac{1}{\sqrt{\frac{e^2}{\epsilon_0} D(E_F)}}$, where k_{TF} is the Thomas-Fermi wave vector and $D(E_F)$ is the electronic density of states (DOS) at the Fermi energy (E_F). In contrast to typical metals ($\lambda_{FT} \sim 0.1$ nm), the calculated

screening length, corresponding to $\delta_{\text{Biz}} \approx 6$ pm observed in the experiments, is $\lambda_{FT} = 0.41 \sim 0.49$ nm for Bi_2Te_3 and $\lambda_{FT} = 0.28 \sim 0.29$ nm for Bi_8Te_9 , both of which decreases with increasing of the polarity amplitude (δ_{Biz}) (Figure S15). Since the screening length is larger than the Bi-Te interlayer distances, the insufficient carrier screening, failing to immediately quench the local polar order [49], also supports the existence of a polar metal state.

One point worth noting is that the weak vdW force is sufficient to establish interlayer dipolar correlation in the 2D materials. This is supported by its analogy to the wavy polarization order observed at charged domain walls of ferroelectric $\text{Pb}(\text{Zr}_{1-x}\text{Ti}_x)\text{O}_3$ and BiFeO_3 [50, 51], and at charged Bi_2O_2 -sheet-linked $\text{Bi}_3(\text{Fe},\text{Ti})_3\text{O}_{13}$ layers [52]. Specifically, as monitoring the in-plane lattice parameter change, we find that the a axis is compressed by -1.44% and expanded by 0.88% near the $\text{Bi}_2\text{Te}_3/\text{Si}$ and $\text{Bi}_8\text{Te}_9/\text{Si}$ interfaces, respectively (Figure S16). Being consistent with the enhanced interfacial coupling, for example, bilayer graphene on SiC and monolayer WSe_2 on black phosphorus [52–55], this unveils that the electrostatic and mechanical interactions are especially strong at the Bi-Te/Si interfaces. The results indicate that the interfacial interaction should play important roles in mediating the n-to-p conduction transition and other exotic phenomena observed in the TIs.

In conclusion, we carry out a systematic electron microscopy study on the Bi-Te thin films and unveil that the intrinsic antisite defects can give rise to exotic polarization orders in the $(\text{Bi}_2)_m(\text{Bi}_2\text{Te}_3)_n$ system, which is supported by SHG measurement results. For the Bi_2Te_3 with random antisite defects, a quasi-cycloidal polarization order is observed along normal direction of the QLs. While for the Bi_8Te_9 with nearly ordered antisite defects, a hyperbolic polarization wave is observed along normal

direction of the QLs. Apart from the face-shared unusual packing of the acentric octahedra, our first-principles calculations reveal that the coexistence of wavy polarization orders with the metallic conductivity is attributed to the insufficient carrier screening in the Bi-Te system. Overall, our findings point out that the defect engineering can be used to manipulate the 2D material property, which not only deepens our understanding on the fundamental property of layered TIs, but also provides a new degree of freedom for future device design.

8 | Methods

8.1 | Sample Growth

Before growth of Bi-Te thin films using molecular-beam epitaxy (MBE), we firstly treated the $10 \times 10 \text{ mm}^2$ Si (111) surfaces by using a standard set of wafer cleaning steps (RCA-HF) to remove the organic contaminations and native oxide. A consecutive HF dip passivates the Si surfaces with hydrogen for the transfer into the MBE chamber (base pressure 5×10^{-10} mbar). To desorb the hydrogen from the surface, the substrates were heated up to 700°C for 10 min and finally cooled down to 275°C and 300°C for growth of Bi_2Te_3 and Bi_8Te_9 , respectively. For evaporation of Te and Bi, standard effusion cells were heated to $T_{\text{Te}} = 330^\circ\text{C}$ and $T_{\text{Bi}} = 460^\circ\text{C}$ for growth of Bi_2Te_3 films. For Bi_8Te_9 , the effusion cells were heated to $T_{\text{Te}} = 285^\circ\text{C}$ and $T_{\text{Bi}} = 480^\circ\text{C}$, respectively. The Te shutter was opened several seconds in advance to saturate the dangling bonds of on the silicon surface.

8.2 | Scanning Transmission Electron Microscopy

To carry out electron microscopy observations, we cut cross-sectional lamella specimens by using FEI Helios 600i focus ion beam (FIB) system. During the milling stage, the Ga-ion beam was adjusted to thin down the lamella specimen, from the initial 30 kV at 790 pA to 5 kV at 68 pA. Fischione plasma cleaner model 1020 was used to remove the surface contamination layers before the STEM experiment. A FEI Titan G² 80–200 ChemiSTEM microscope equipped with a high-brightness Schottky field emission electron gun, a Cs probe corrector, annular detectors and a Super-X energy-dispersive X-ray spectrometer was used to characterize the samples. The electron beam convergence semi-angle is 24.7 mrad and the collection semi-angle is in the range of $70\text{--}200 \text{ mrad}$. Dr. Probe software package was used to simulate the HAADF-STEM images and 2D Gaussian peak fitting based on maximum likelihood estimation was employed to measure the atomic positions. The measured lattice parameters of the Bi-Te phases are calibrated by referring to that of the Si substrate.

8.3 | Octahedral Polarity Calculation

In QL-stacked Bi_2Te_3 , the Te1 atoms are threefold coordinated while the Te2 atoms are sixfold coordinated due to their face-sharing feature in the octahedral pair. On this basis, one can get that the negative charge center of an octahedron along z axis. By combining $\delta_i - \delta_0 = 0$ from and the electroneutral law,

$$\nu + \mu_1 + 0.5 \mu_2 = 0 \quad (3)$$

one can get that

$$\mu_1 = \frac{\mu_2 (z_{\text{Bi}} - z_{\text{Te2}})}{2(z_{\text{Te1}} - z_{\text{Bi}})} \quad (4)$$

$$\mu_1 = 0.5947 \mu_2 \quad (5)$$

here, the variables for the specific octahedron are $z_{\text{Bi}} = 0.40050$, $z_{\text{Te1}} = 0.45697$, and $z_{\text{Te2}} = 0.33333$, respectively [39].

8.4 | First-Principles Calculations

Density functional theory (DFT) calculations are performed using the projector augmented-wave (PAW) method implemented in the VASP code. The exchange and correlation energies were treated with the generalized gradient approximation (GGA) parameterized by the Perdew–Burke–Ernzerhof formula. PAW potentials with valence electronic configurations $5d^{10}6s^26p^3$ and $5s^25p^4$ are used for the Bi and Te atoms, respectively. The energy convergence criterion is 10^{-5} eV , and the ionic relaxations stopped when the forces on every atom became smaller than $0.02 \text{ eV/\AA}$. A plane-wave basis set was used with a kinetic energy cutoff of 300 eV . An $8 \times 8 \times 1$ k -point mesh was used for Bi_2Te_3 bulk model (20 atoms) and slab model (30 atoms). An $8 \times 8 \times 2$ k -point mesh was used for Bi_8Te_9 bulk and slab model (51 atoms). These settings are found to be sufficient for the convergence of the density of state and band structure. The vdW corrections were included using the DFT-D3 method with Becke–Johnson damping. Given the layered nature of the materials, vdW interactions have a significant effect on the interlayer spacing. We tested different vdW correction schemes and found that DFT-D3 with Becke–Johnson damping yields the most reasonable lattice parameters for Bi_2Te_3 compared to experiments (Table S1). The spin-orbit coupling (SOC) was included in the band structure calculations.

Author Contributions

X.-K.W. conceived the idea, collected and analyzed the data and coordinated the study. A.R.J. grew the samples. J.J. performed the first-principles calculations under the guidance of Y.S. M.W. contributed helpful discussions on DFT calculations. Q.L. and L.Y. carried out the SHG experiments. X.-K.W. wrote the manuscript, all authors discussed the results and revised the manuscript.

Acknowledgements

We would like to thank Prof. Joachim Mayer for his contribution in project application, supervision and helpful discussion to this work. This project was funded by the Deutsche Forschungsgemeinschaft (DFG, German Research Foundation) under Germany's Excellence Strategy—Cluster of Excellence Matter and Light for Quantum Computing (ML4Q) EXC 2004/1 – 390534769. X.-K.W. thanks the financial support from the National High-Level Youth Talents Program (Grant Nos. 0040/X2450224, 0040/K2923004), the National Science Foundation of China (Grant Number. 52573256), the Xiamen Natural Science Foundation Program (Grant Number. 3502Z202573024), the Xiamen University President's Fund Program (Grant Number. 20720230006) and the State Key Laboratory of Vaccines for Infectious Diseases of Xiang An Biomedicine Laboratory (Grant Number. 2024XAKJ0100006). Y.S. and J.J. were supported by the National Natural Science Foundation of China (Grant No. T2422016).

Funding

This project was funded by the Deutsche Forschungsgemeinschaft (DFG, German Research Foundation) under Germany's Excellence Strategy—Cluster of Excellence Matter and Light for Quantum Computing (ML4Q) EXC 2004/1 – 390534769. X.-K.W. thanks the financial support from the National High-Level Youth Talents Program (Grant Nos. 0040/X2450224, 0040/K2923004), the National Science Foundation of China (Grant No. 52573256), the Xiamen Natural Science Foundation Program (Grant No. 3502Z202573024), the Xiamen University President's Fund Program (Grant No. 20720230006) and the State Key Laboratory of Vaccines for Infectious Diseases of Xiang An Biomedicine Laboratory (Grant No. 2024XAKJ0100006). Y.S. and J.J. were supported by the National Natural Science Foundation of China (Grant No. T2422016).

Conflicts of Interest

The authors declare no competing interests.

Data Availability Statement

The data that support the findings of this study are available on request from the corresponding author. The data are not publicly available due to privacy or ethical restrictions.

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Supporting Information

Additional supporting information can be found online in the Supporting Information section.

Supporting File: adma74576-sup-0001-SuppMat.pdf.

Intrinsic Defect-Induced Exotic Polarization Waves in Van der Waals Layered Topological Insulators

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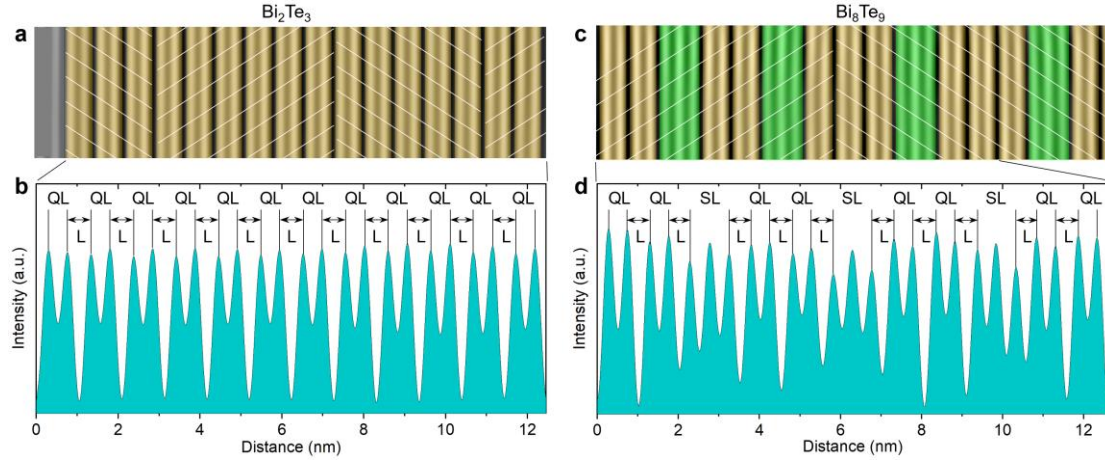


Figure S1. Identification of QLs and SLs in the two Bi-Te samples. (a,b) IFFT image of the Bi_2Te_3 phase and corresponding intensity profile showing the modulation spacing between the QLs, respectively. The vdW gap marked by L is around 0.583 nm and in between it is 0.463 nm. (c,d) IFFT image of the Bi_8Te_9 phase and corresponding intensity profile showing the modulation spacing between the QLs and SLs, respectively. The vdW gap marked by L is around 0.557 nm and in between it is 0.460 nm.

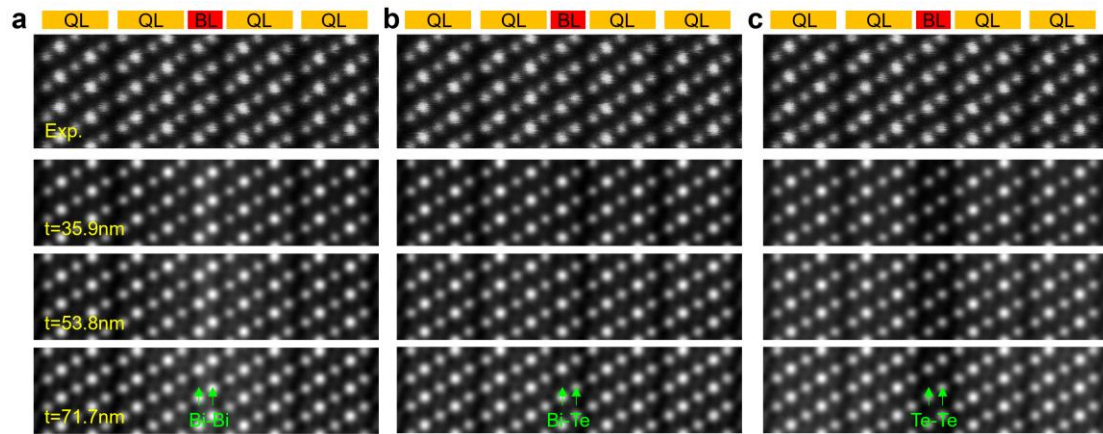


Figure S2. Comparison of experimental with simulated HAADF STEM images about the BL sandwiched by two QLs. (a-c) The QL-BL-QL stacking is simulated based on multi-slice method at different thickness (horizontal rows for $t = 35.9$, 53.8 and 71.7 nm) with the Bi-Bi, Bi-Te and Te-Te type BL structural models, respectively.

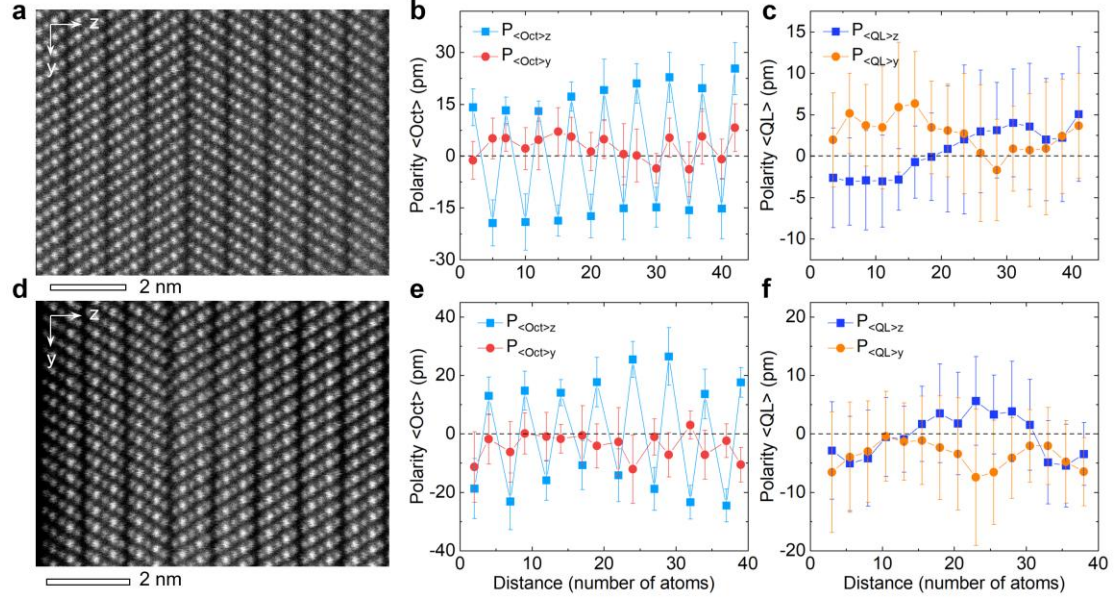


Figure S3. Reproducible wavy-like polar order in Bi_2Te_3 thin films. (a-c) HAADF-STEM image collected along $[100]$ direction and polarity profiles of #1 (a-c) and #2 (d-f) sample for the octahedral units and QLs plotted as a function of distance along z direction, respectively. One observes the polarity reversal along z direction not only near the twin-like boundary, but also inside the ideal structural domain (f).

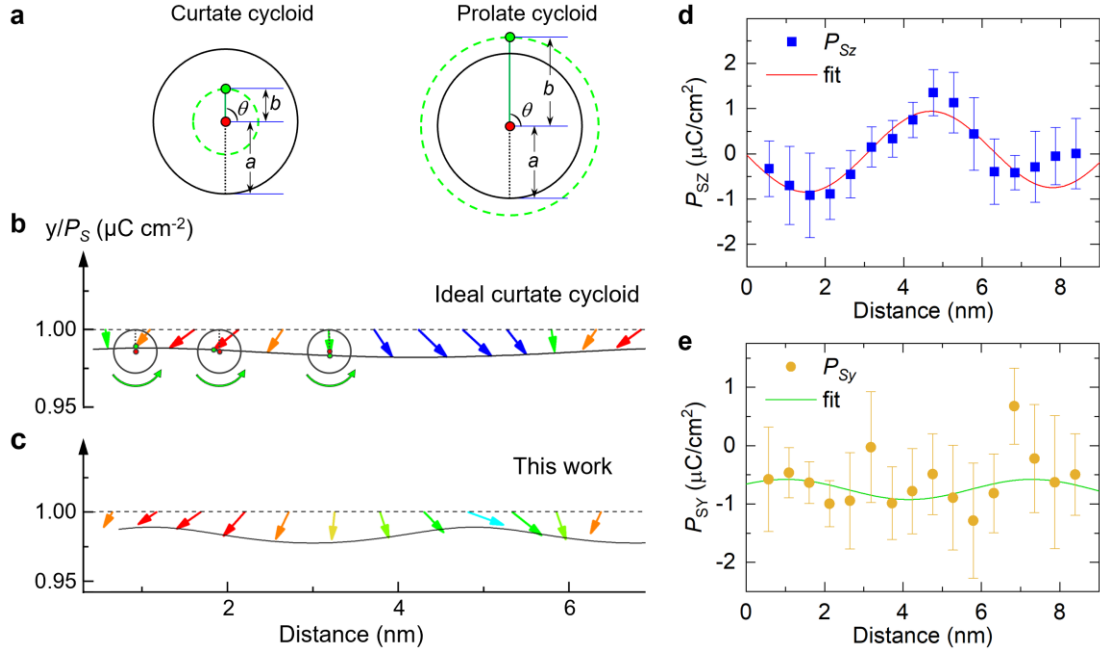


Figure S4. Quasi-cycloidal polarization order in Bi_2Te_3 films. (a) Illustration of the curtate and prolate cycloid models. (b) An ideal curtate cycloid with presentation of vector orientation change in 2D space. (c) The 2D polarization variation observed in our Bi_2Te_3 film (Figure 3e), which can be nicely described by a wavy order closely approximate to the curtate cycloid. (d,e) Mathematical fitting to the P_{Sz} and P_{Sy} profiles by using $z = a_1 * x - b_1 * \sin(x + c_1)$ for (d) and $y = a_2 - b_2 * \cos(x + c_2)$ for (e), respectively. The best-fit parameters are $a_1 = 0.015$, $b_1 = 0.869$, $c_1 = 0.037$ and $a_2 = -0.751$, $b_2 = -0.173$, $c_2 = -1.007$.

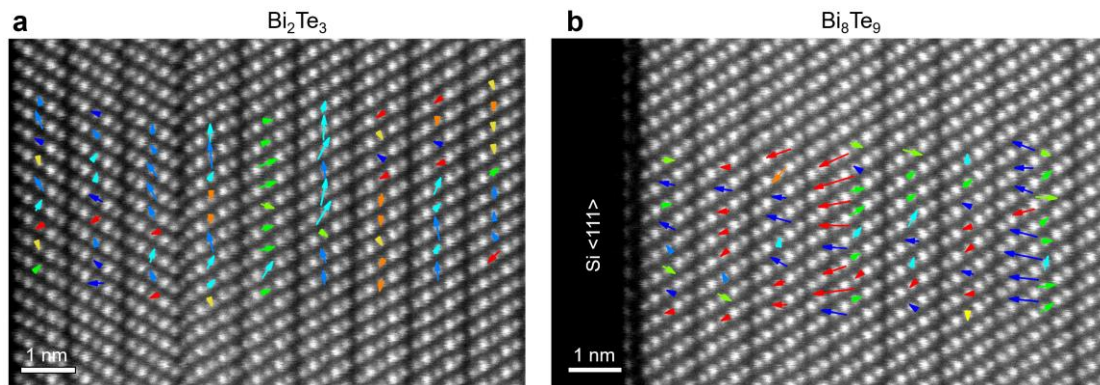


Figure S5. Overlapping of QL polarity on the HAADF images. (a,b) The polarity vectors overlapped on the Bi₂Te₃ phase and Bi₈Te₉ phase, respectively. The arrow colour highlights orientation of the polarization vectors.

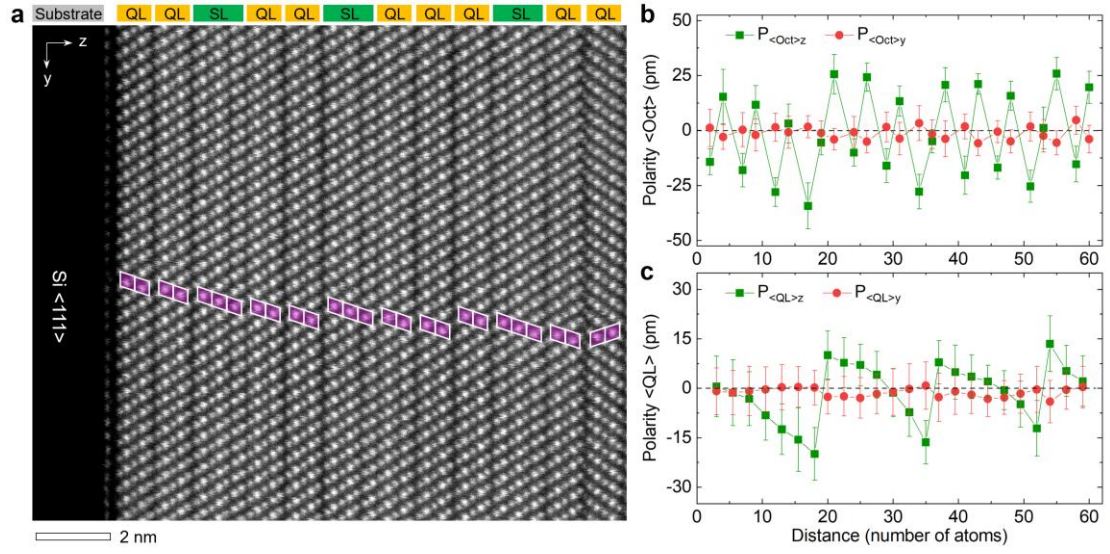


Figure S6. Depth-dependent evolution of hyperbolic polar wavy in Bi₈Te₉ thin film. (a) HAADF-STEM image collected along [100] direction. The QL and SL layers are annotated. (b,c) Polarity profiles for the octahedral units and QLs plotted as a function of distance along z direction, respectively.

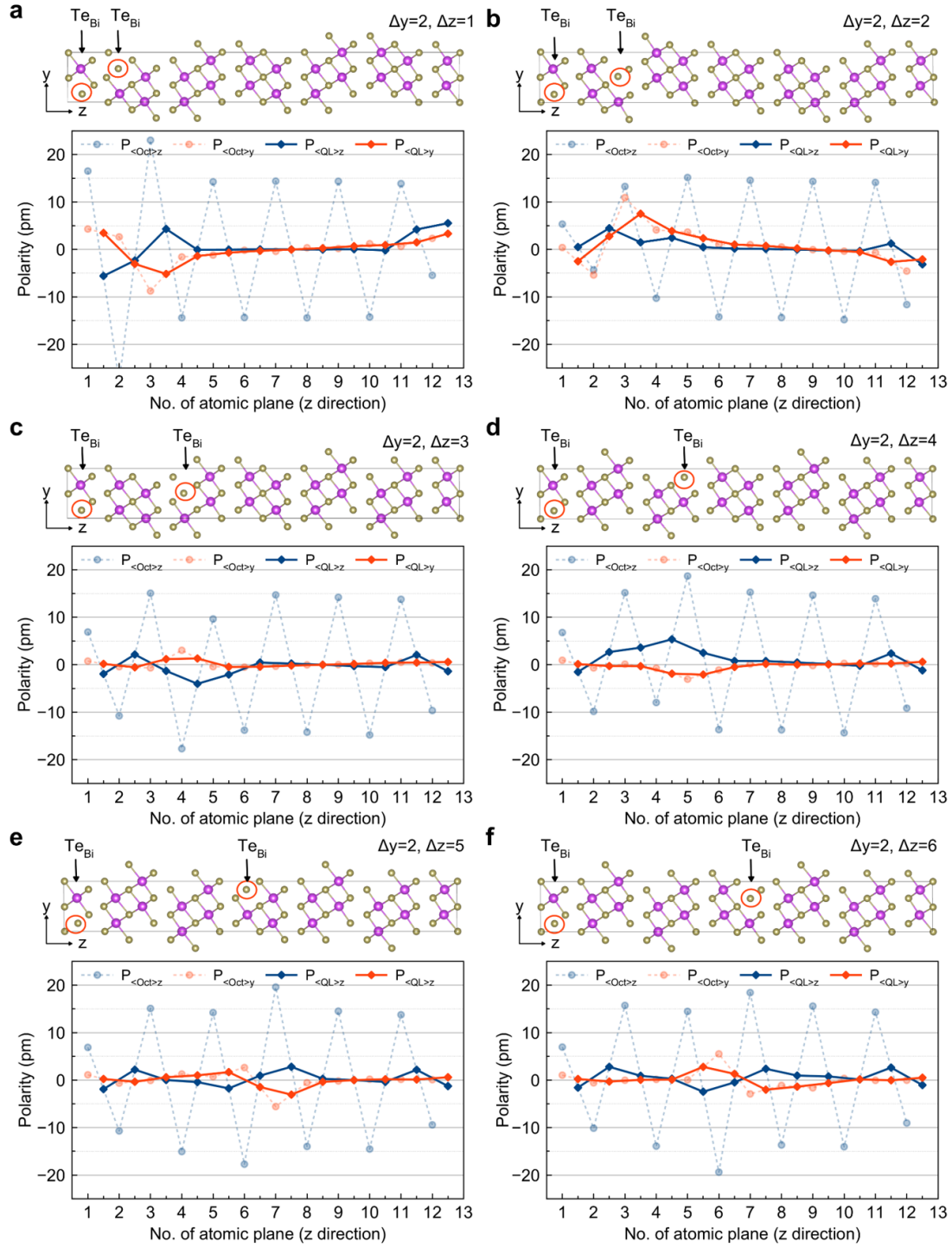


Figure S7. Single-site Te_{Bi} antisite defects induced 3D polarity and long-range dipolar correlation in Bi_2Te_3 supercells. (a-f) Supercell structural models showing location of the Te_{Bi} defect pairs in octahedral units with $\Delta y = 2$ and $\Delta z = 1 \sim 6$ (upper) and polar displacements of Bi atoms projected along y and z direction in the octahedra (Oct) and QLs (lower), respectively.

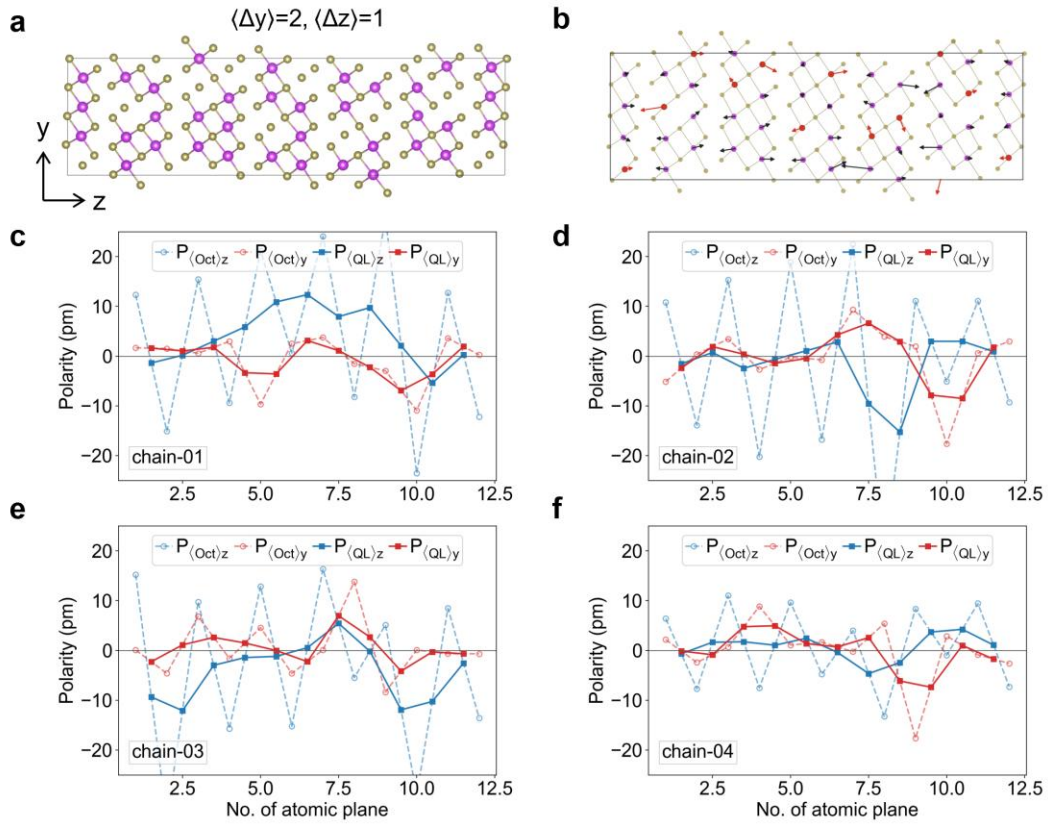


Figure S8. High-density TeBi-type antisite defect induced polarity in Bi₂Te₃. (a) Schematic supercell with an average TeBi- TeBi spacing of $\langle \Delta y \rangle = 2$ and $\langle \Delta z \rangle = 1$ viewed along [100] direction. (b) Mapping of polar Bi shifts in the octahedra. (c-f) Polar Bi-shift profiles along y and z direction for horizontal octahedral chains, respectively. The chain-01 ~ chain-04 are numbered from the top to the bottom row.

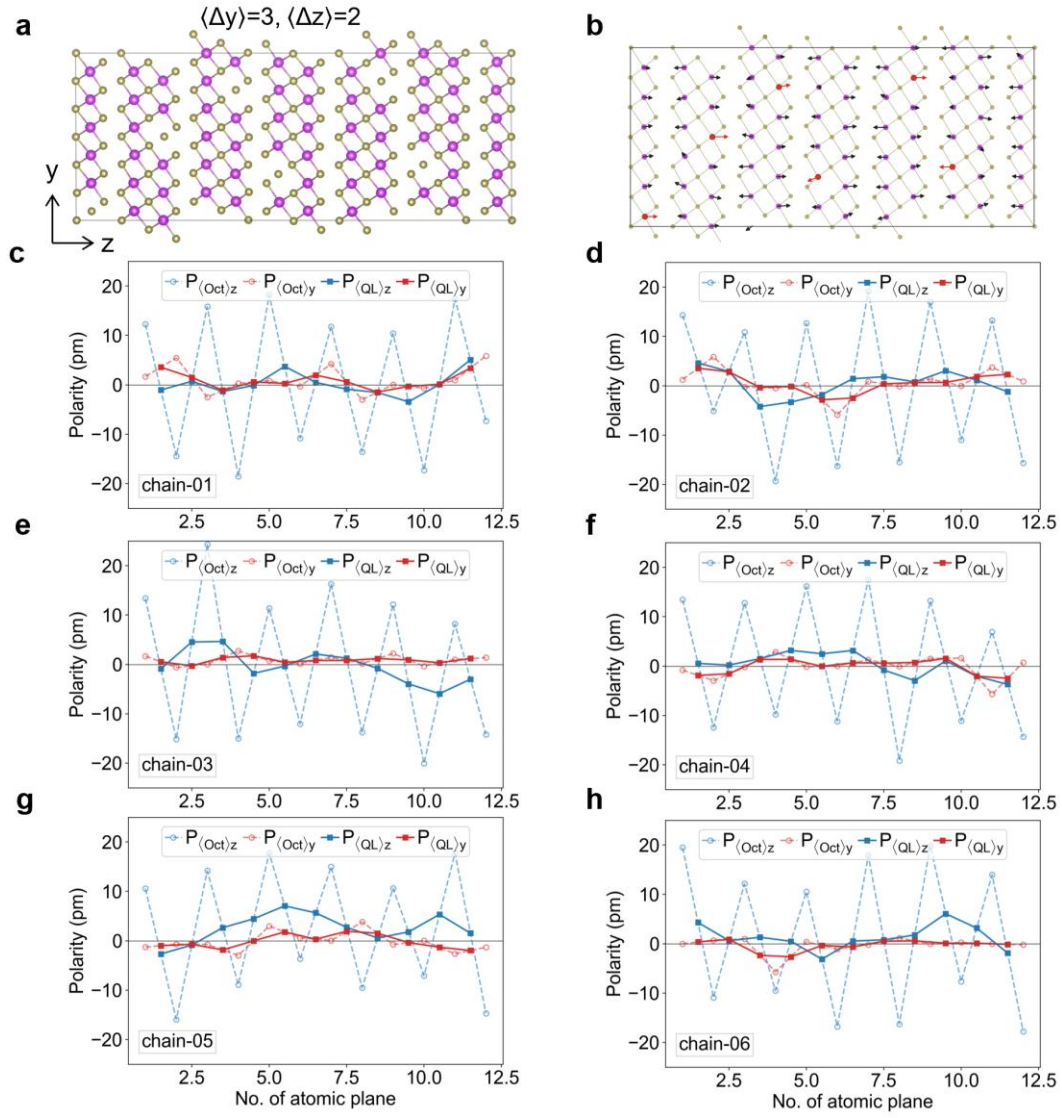


Figure S9. Medium-density TeBi -type antisite defect induced polarity in Bi_2Te_3 . (a) Schematic supercell with an average TeBi - TeBi spacing of $\langle \Delta y \rangle = 3$ and $\langle \Delta z \rangle = 2$. (b) Mapping of polar Bi shifts in the octahedra. (c-h) Polar Bi-shift profiles along y and z direction for horizontal octahedral chains, respectively. The chain-01 ~ chain-06 are numbered from the top to the bottom row.

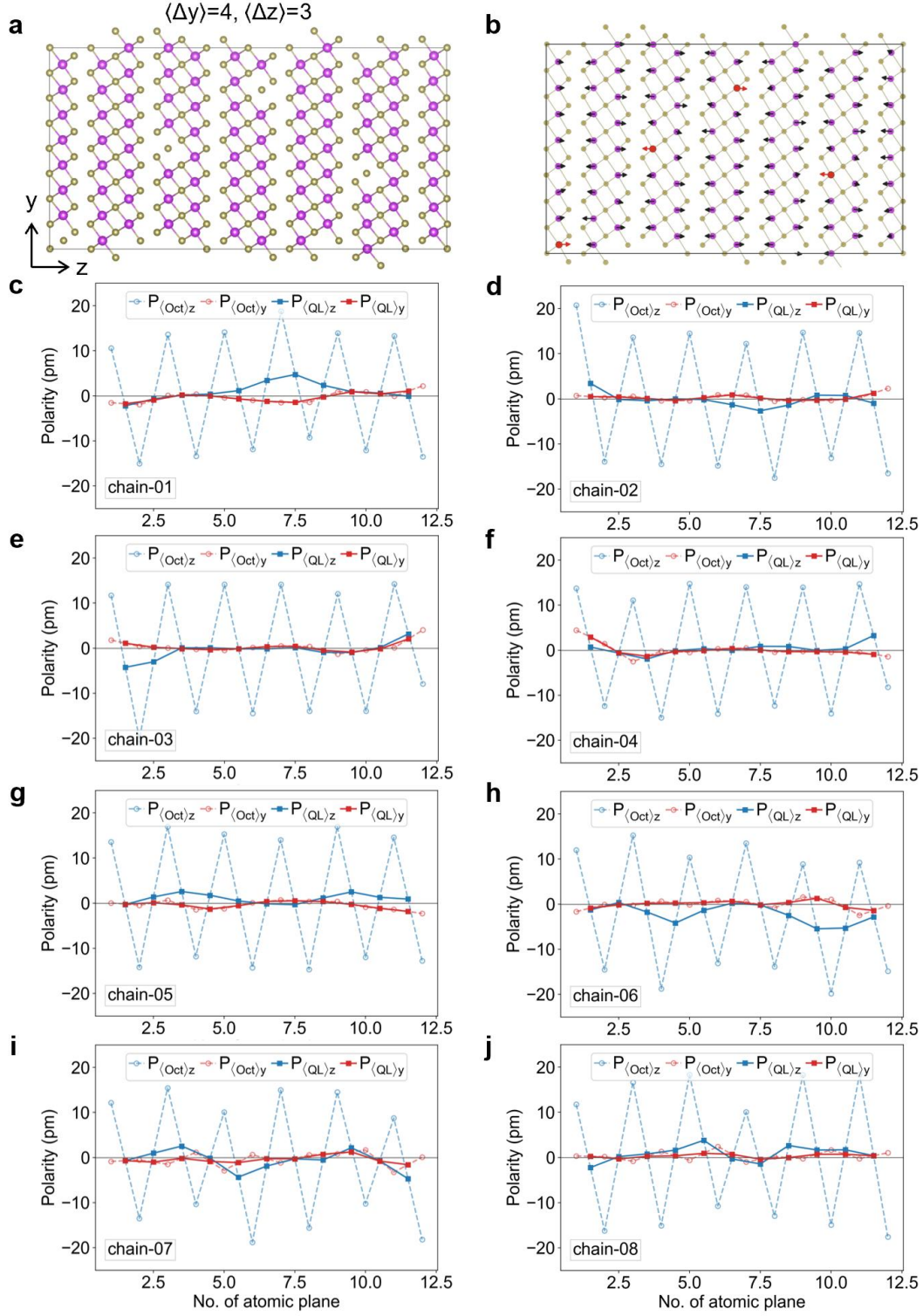


Figure S10. Low-density TeBi -type antisite defect induced polarity in Bi_2Te_3 . (a) Schematic supercell with an average TeBi - TeBi spacing of $\langle \Delta y \rangle = 4$ and $\langle \Delta z \rangle = 3$. (b) Mapping of polar Bi shifts in the octahedra. (c-f) Representative polar Bi-shift profiles along y and z direction for horizontal octahedral chains, respectively. The chain-01 ~ chain-04 are numbered from the top to the bottom row.

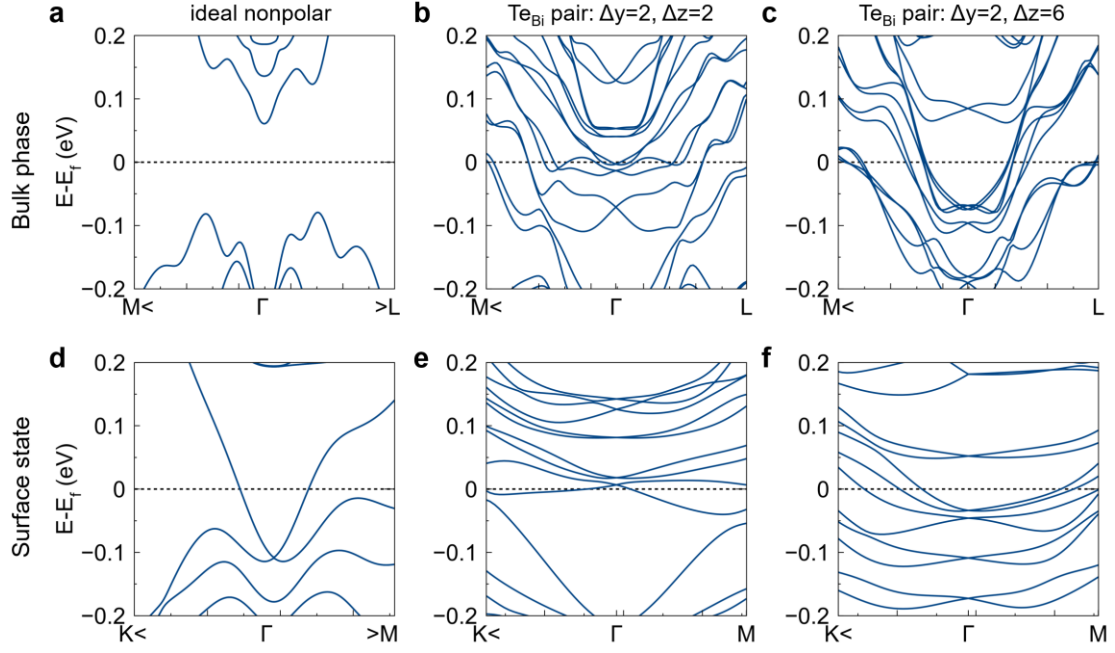


Figure S11. Comparison of bulk-phase and surface-state electronic structures in Bi_2Te_3 . (a-c) Bulk-phase electronic band structures of ideally non-polarized Bi_2Te_3 , supercells containing a pair of Te_{Bi} antisite defects with ($\Delta y=2$, $\Delta z=2$) spacing and ($\Delta y=2$, $\Delta z=6$) spacing, respectively. (d-f) Surface state electronic band structures of ideally non-polarized Bi_2Te_3 , supercells containing a pair of Te_{Bi} antisite defects ($\Delta y=2$, $\Delta z=2$) spacing and ($\Delta y=2$, $\Delta z=6$) spacing, respectively. The first-principles calculations are carried out with the SOC strength of $f_{\text{SOC}}=1.0$.

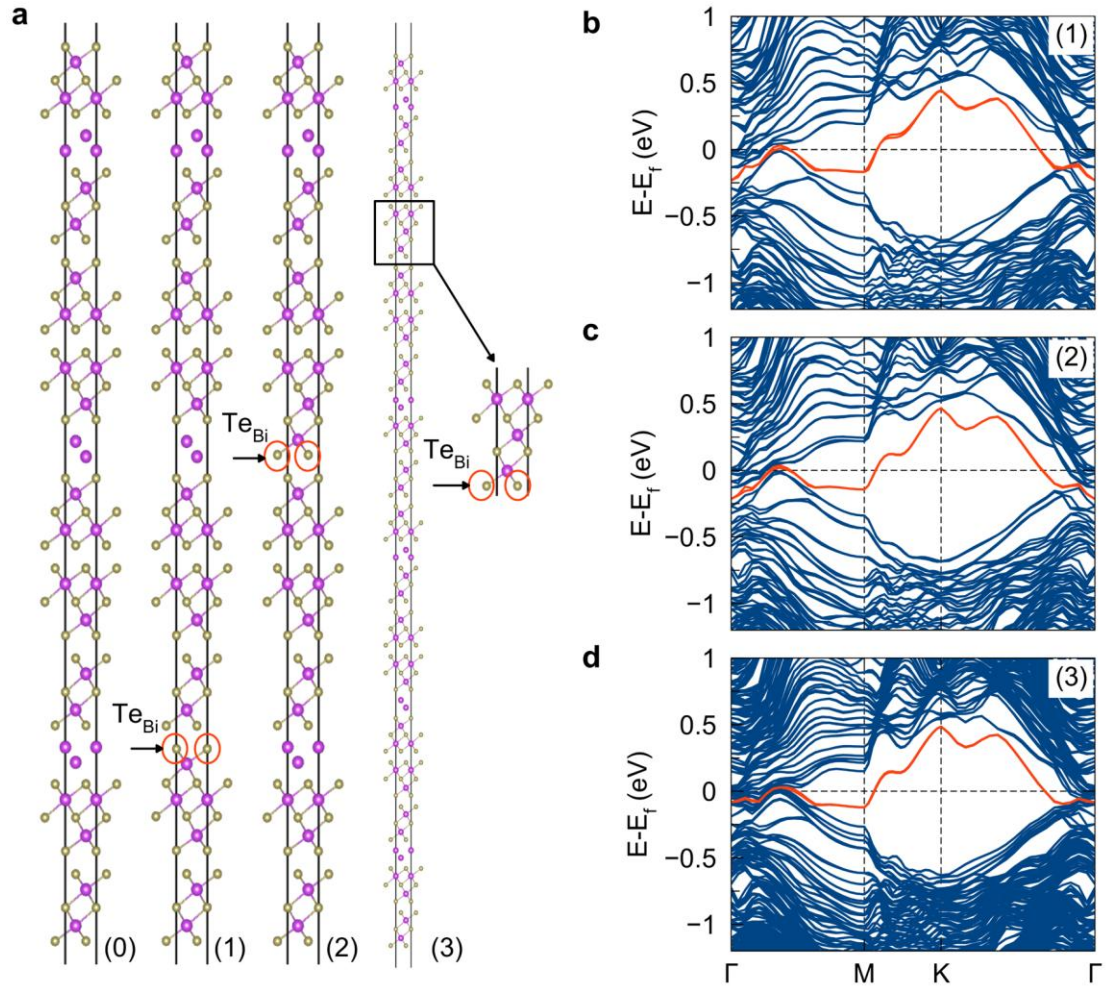


Figure S12. Modification of surface electronic band structures by Te_{Bi} antisite defects in Bi₈Te₉ phase. (a) Slab models separated by a 3 nm vacuum layer: (0) A pristine 3 f.u. supercell; (1) and (2) single-array Te_{Bi} antisite defects in 3 f.u. supercells located near the surface and in the interior, respectively; (3) two arrays of Te_{Bi} antisite defects in a 3 f.u. supercell; (4) single-array Te_{Bi} antisite defects in a 6 f.u. supercell. The atom types are Bi-purple, Te-gold. (b) Electronic band structures corresponding to the structural models of (1)-(3).

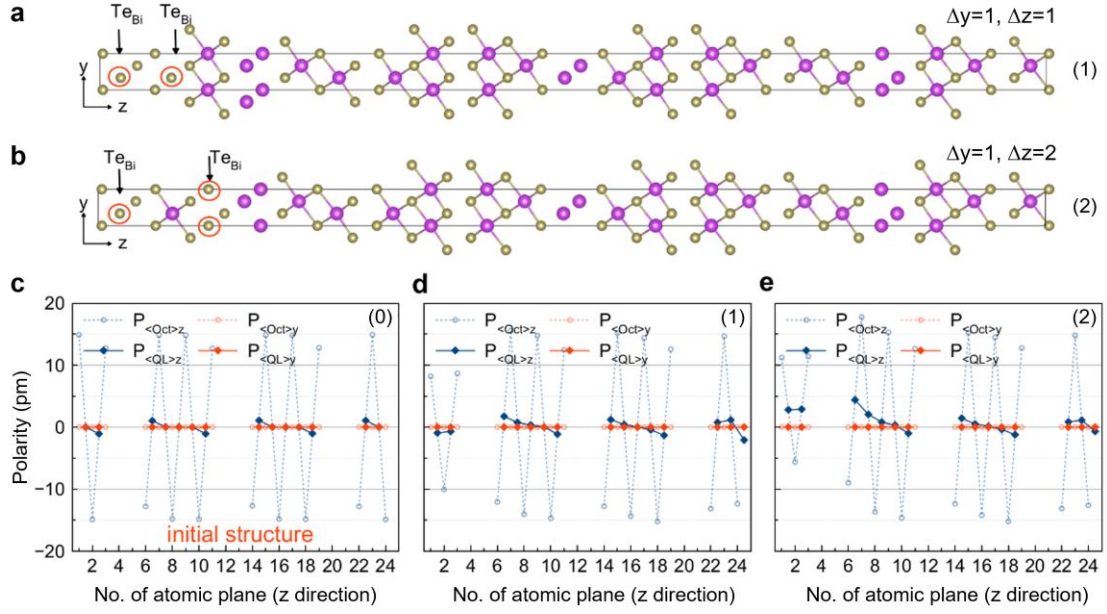


Figure S13. The Te_{Bi}-type antisite defect induced quasi-1D polarity feature in Bi₈Te₉. (a,b) Illustration of two-array Te_{Bi}-type antisite defects with a spacing of $(\Delta y = 1, \Delta z = 1)$ for (a)-(1) and $(\Delta y = 1, \Delta z = 2)$ for (b)-(2) in the projected yz plane, respectively. (c-e) Polar displacements of Bi atoms along y and z direction in the octahedra (Bi-Oct) and QLs (average-Oct) for the ideal initial structure (0) and that with antisite defects of scenario (1) and (2), respectively.

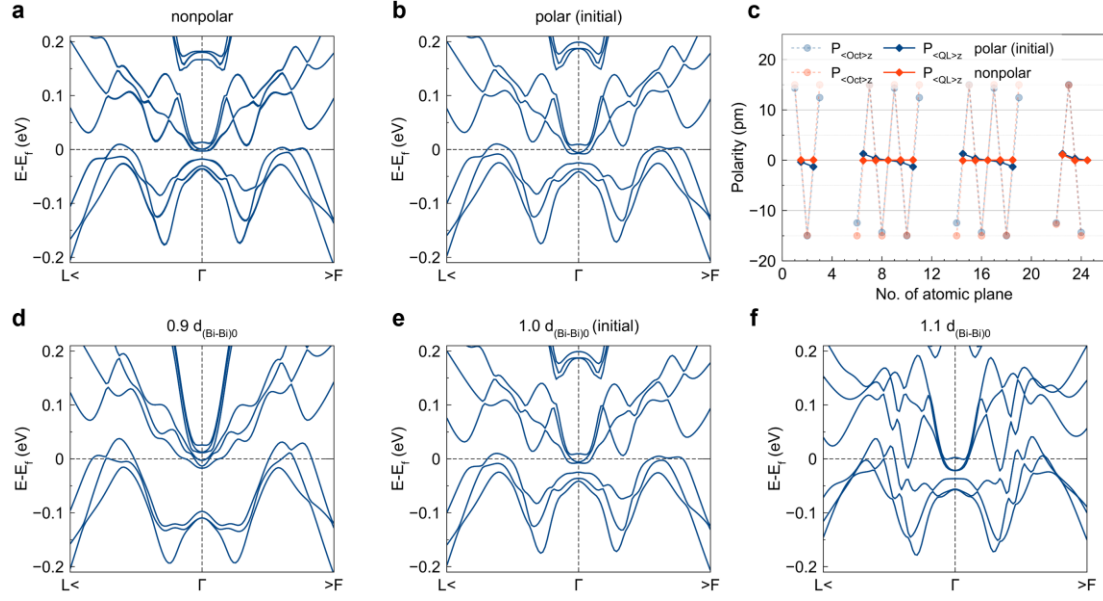


Figure S14. Polarity and Bi-Bi intralayer spacing induced band structure change in Bi_8Te_9 phase. (a,b) Band structures for the nonpolar and polar ($P_{\langle \text{QL} \rangle z} = 1.2$ pm) models, respectively. (c) Polarity profiles for the nonpolar and polar structural models. (d-f) Bi-Bi intralayer spacing induced band structure change, with $d_{\text{Bi-Bi}} = 0.9, 1.0$ and $1.1 d_{\text{Bi-Bi}0}$, $d_{\text{Bi-Bi}0} = 0.169$ nm.

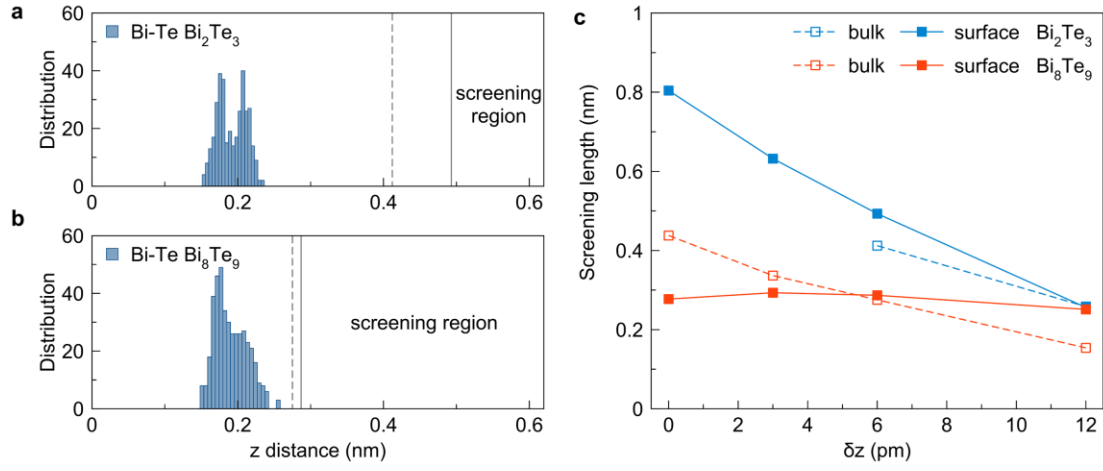


Figure S15. Experimentally measured Bi-Te interlayer-distance distributions along the z direction for (a) Bi_2Te_3 and (b) Bi_8Te_9 . (c) first-principles calculated Thomas-Fermi screening length as a function of the QL polarity (δz) for Bi_2Te_3 (blue) and Bi_8Te_9 (red). Dashed lines denote bulk phases, and solid lines denote surface phases.

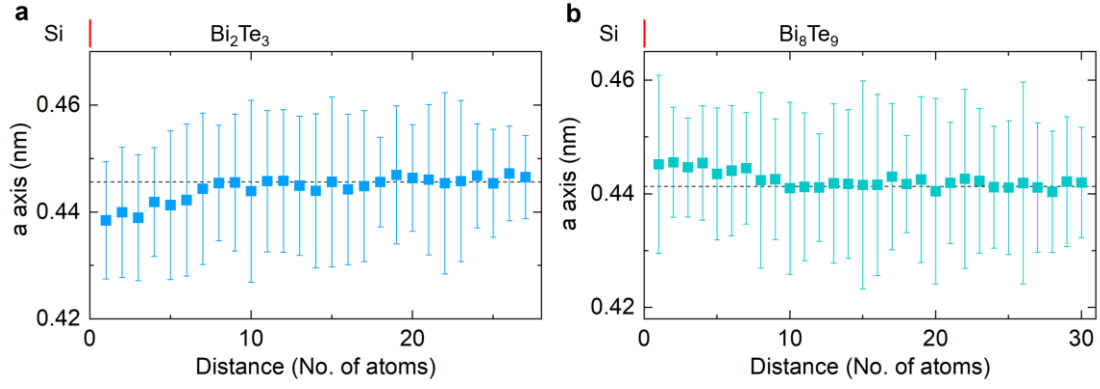


Figure S16. Lattice parameter changes as a function of distance away from the Si/Bi-Te interface. (a,b) The a axis change for the Si/Bi₂Te₃ and Si/Bi₈Te₉ interface, respectively. With respect to the interior mean value of a axis in the Bi-Te film, i.e., 0.4456 nm for the former (a) and 0.4416 nm for the latter (b), the a axis is compressed by -1.44% and expanded by 0.88% near the interface, respectively.

Table S1. Lattice constants of the Bi_2Te_3 from experiments and different DFT methods.

Lattice constant	Experiment	no correction	DFT-D3 method of Grimme with zero- damping function	DFT-D3 method with Becke-Johnson damping function
a(Å)	4.385	4.445	4.430	4.383
c(Å)	30.497	31.688	30.234	30.132